Internal division of a segment

1 Positive

2

a $x = \dfrac{mx_2 + nx_1}{m+n}$ **b** $y = \dfrac{my_2 + ny_1}{m+n}$

3 $\left(\dfrac{9}{4}, \dfrac{47}{4}\right)$

4

a $(2,4)$ **b** $(0,5)$ **c** $\left(\dfrac{1}{7}, \dfrac{74}{7}\right)$

5

a $\left(\dfrac{24}{5}, 4\right)$ **b** $\dfrac{9}{5}$

6

a $\left(\dfrac{14}{3}, \dfrac{7}{3}\right)$ **b** $\left(\dfrac{19}{3}, \dfrac{13}{2}\right)$ **c** $\dfrac{5}{2}$ **d** $\dfrac{5}{2}$

e Yes

External division of a segment

7 Negative

8

a $(6,21)$ **b** $\left(\dfrac{5}{2}, -\dfrac{1}{2}\right)$

9 $(1,8)$

10

a $1:5$ **b** $6:-5$ **c** $-1:6$

11 $(2,4)$

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